Meaningful Comparisons with Likert Scales

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This paper is written in R Markdown which includes all code for analyses and figures. The R Markdown file and all supporting files are curated at https://github.com/PerceptionAndCognitionLab/bf-likert. The accompanying web applet is avilable at https://martinschnuerch.shinyapps.io/likertBF/; underlying source code is avilable at https://github.com/mschnuerch/likertBF.

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Abstract

Although Likert rating scales are exceedingly popular in psychological research, there is often a lingering question of whether metric-level descriptions of the results and parameteric tests are appropriate. We consider a different problem, perhaps one that supersedes the ordinal-vs-metric issue: When is it appropriate to use single relations—say, a difference in central tendencies—to compare two Likert-item distributions? In this paper, we propose a Bayesian approach to guide researchers in drawing meaningful inferences from Likert items. We develop four statistical models that represent possible relations across two ratings distributions: a null model, a parametric singel-relation model, a nonparametric single-relation model, and a multiple-relations model that accounts for complex effects. We show how these models can be compared in light of data with Bayes factors and illustrate their usefulness with two real-world examples. We also provide a freely available web applet for researchers who wish to adopt the approach.

Keywords: Likert scales, Likert items, Meaningfulness, Bayes Factors, Stochastic Dominance

Meaningful Comparisons with Likert Scales

It is hard to overstate the popularity of Likert scales in social science research. Since their introduction by Rensis Likert (1932), their application has become common practice in political science, economics, educational research, policy research, health sciences, and psychology. Although the construction of a Likert scale often refers to the summation of responses across several question, here we focus on the foundation of this summation—the Likert question or item itself (see Clason & Dormody, 1994). The key question is how responses on a Likert item are affected by covariates, say condition factors or subject factors. Almost always, the comparison across covariates is characterized with means and standard deviations and formally follows from t-tests. But are means and standard deviations justified? And do t-tests allow for meaningful comparisons?

Consider the following hypothetical example: Suppose we wanted to compare the sadness of 100 first-year Marines and 100 first-year college students. Each individual indicated on a 5-point Likert item how often they felt sad, with response options ranging from "never" to "always". Table 1 shows hypothetical data for two different scenarios labeled plainly Scenario I and Scenario II. Let's consider Scenario I first: Following the convention, we calculate mean responses of 2.50 and 3.00 for the Marines and college students, respectively. Thus, for this hypothetical example, Marines exhibit substantially less sadness than college students, $t(198) = 2.57$.

Although the above analyses are nearly universal, there is also near universal recognition that Likert scales are ordinal scales rather than interval scales (Clason & Dormody, 1994; Göb, McCollin, & Ramalhoto, 2007; Jamieson, 2004). And if we stay true to an ordinal interpretation, metric quantities such as means, standard deviations, and t-values are not meaningful. Perhaps this critique is captured most eloquently by Sullivan and Artino Jr (2013) who ask, "What does the average of *Never* and *Rarely* really mean?" (italics added, p. 541) Even so, while researchers are aware of this critique, it does not prevent them from

calculating means or performing parametric tests. We all follow the convention and carry on.

Of course, we are not the first to consider this issue. Some have defended the use of parametric statistics with reference to their robustness against violations of underlying assumptions (e.g., Norman, 2010). When researchers question the appropriateness of metric statistics, however, the usual course is to use ordinal or nonparameteric statistics. For example, Nanna and Sawilowsky (1998) compared the t-test to a nonparametric Wilcoxon rank-sum test on error control and statistical power. They found that the nonparametric outperformed the parametric test in almost all scenarios examined. Their recommendation was thus to use nonparametric tests to analyze Likert-scale data. The very same recommendation can be found in many fields. For example, in biomedical research, Jamieson (2004) states that "nonparametric tests should be employed if the data is clearly ordinal" (p. 1218). Kuzon, Urbanchek, and McCabe (1996) go even further in declaring the use of parametric statistics on ordinal data as one of the "Seven Deadly Sins of Statistical Analysis". Yet, we think these statements and proposed solutions do not address the heart of the problem.

The critical question is, "*How can we describe the comparison of two conditions when using Likert items?*" There are two obvious answers in the literature. The first is the null relation—there is no true difference between the two conditions once we account for sampling variability. The second is the effect relation—there is some true difference *and* it is captured by a single effect-size measure and corresponding test. Consider Scenario I in Table 1. Here, there is a single relation—college students are unambiguously sadder than Marines, and this relation holds across the scale.

Yet, there is another position where there is an effect that is not captured by a single relation (Clason & Dormody, 1994). Consider Scenario II. If we only looked at the means, we would find that there is no difference between Marines and college students (both have a mean value of 3.00). Looking at the distributions across the scale, however, this conclusion

hardly seems justified. Obviously, the effect of being in the Marines is polarizing: While many Marines report *never* being sad, many also report *always* being sad. No single relation captures this polarizing result. Marines are seemingly both more happy and more sad than their college peers. Any characterization of a single relation, say a mean or a median, misses this structure. Hence, only looking at the location is not meaningful for these data. And tests comparing locations, even nonparametric tests, are also not meaningful (Clason & Dormody, 1994).

The above example shows that the patterns in the data themselves lead to different approaches to comparing conditions. For Scenario I, tests on a single relation capture the structure in the data; for Scenario II, in contrast, they do not. In this paper, we provide a Bayesian approach to help researchers use data to draw appropriate inferences from Likert items. In the following, we develop four statistical models that represent possible, meaningful relations across two Likert-item distributions: a null model, a parametric single-relation model, a nonparametric single-relation model, and a multiple-relations model to account for complex effects like polarization. We show how these models may be evaluated in light of data by means of Bayes factors and present a user-friendly web applet for readers who wish to adopt the analysis in their own research. Finally, we demonstrate the usefulness of the approach by applying it to two real-world examples, and assess the sensitivity of Bayes factor model comparisons to reasonable variations in prior settings.

Bayesian Models for Likert Scale Data

We use four statistical models to compare Likert-item distributions across conditions. The base of the models is a signal-detection-like setup, and the upper-left panel of Figure 1 shows the construction in the context of our hypothetical example. There are latent ratings—assumed to be a draw from a standard normal (mean of 0.0 and standard deviation of 1.0)—which represents the latent continuous attribute underlying responses to the discrete Likert item. The normal is divided into regions by bounds, and these regions correspond to

the Likert scale responses. The probability that an observation is in a certain region is simply the area under the curve.

The parameters in this setup are the bounds themselves. Let γ_{ij} denote the bound between response category j and $j+1$ ($j = 1, \ldots, J$) in condition i ($i = 1, 2$). For the setup to be valid, bounds within each condition have to order, that is,
$\gamma_{i0} = -\infty \leq \gamma_{i1} \leq \ldots \leq \gamma_{iJ} = \infty$.

To make comparisons, we adopt the following regression-like extension. Let α_j and θ_j be averages and differences across conditions as follows. Let $\alpha_j = (\gamma_{1j} + \gamma_{2j})/2$ be the average for the jth bound, and let $\theta_j = \gamma_{2j} - \gamma_{1j}$ be the difference for the jth bound. Combining,

$$\gamma_{ij} = \alpha_j + x_i \theta_j$$

where $x_1 = -1/2$ and $x_2 = 1/2$. The key feature of this parameterization is that θ_j denotes a comparison of the distribution for the jth bound. By placing constraints on θ_j, we can model different types of relations among the two Likert-item distributions.

Statistical Models

We specify four statistical models on θ_j, each representing increasingly more complex relations among the conditions. The models are shown in Figure 1.

Null Model: The first row depicts the null model which posits that there is no effect of condition. This model is denoted $\mathcal{M}_0$, and imposes the constraint $\mathcal{M}_0\colon\ \theta_j = 0$. Thus, the corresponding bounds for college students and Marines are identical. For example, the value of the bound between *Never* and *Rarely* for college students is -0.84, and this value is the same for the bound between *Never* and *Rarely* for Marines. Because all the corresponding bounds are the same in value, the distributions are the same as well. There is no difference among the conditions; hence, there is no effect. The null model has one free parameter for

each bound, that is, $J - 1$ parameters in total. In the hypothetical example displayed in the figure, $J = 5$.

Constant Shift Model: The next row describes a very simple effect where the bounds in one condition all shift by the same amount compared to the other condition. The model is denoted by $\mathcal{M}_1$, and imposes the parametric constraint $\mathcal{M}_1\colon\ \theta_j = \theta^*$. For example, the value of the bound between *Never* and *Rarely* for Marines is -0.54, and this value is 0.30 greater than the bound between *Never* and *Rarely* for college students. This difference is preserved across corresponding bounds. For example, the values of the bounds between *Rarely* and *Sometimes* is 0.05 and -0.25 for Marines and college students, respectively. The difference, 0.30, is the same as between *Never* and *Rarely.* The constant shift model explicitly states that the effect of condition on the ratings can be captured by a single parameter θ^*. In statistics, this model is well known and is the basis for ordinal regression (Albert & Chib, 1993; Fahrmeir & Tutz, 2013; McCullagh & Nelder, 1989). It is comprised of J free parameters (i.e., $J - 1$ means α_j and one difference θ^*).

Dominance Model: The following row shows the dominance model, $\mathcal{M}_d$, which is defined by relaxing the constant shift constraint. For this model, there are a total of $2 \times (J - 1)$ parameters: $J - 1$ mean-bound parameters (α_j) and equally many difference parameters (θ_j). To capture the notion of unambiguity, we impose an order constraint in this model: $\mathcal{M}_d\colon\ \theta_j > 0$. For the example in the third row of Figure 1, γ_{11} is the bound that separates *Rarely* from *Never* for college students, and it has a value of -0.67. Likewise, the value for Marines is denoted γ_{21} and has a value of -0.04. Here, we see that $\gamma_{21} > \gamma_{11}$—Marines have a higher probability of being never sad than college students. Importantly, this inequality holds for all corresponding bounds, that is, because $\theta_j > 0$ for all bounds it follows that $\gamma_{2j} \geq \gamma_{1j}$ for all bound pairs. This property is called *stochastic dominance*, and it means that for any given category, the cumulative proportion in one condition is larger than or equal to the cumulative proportion in the other condition.

There are two possible dominance conditions: one in which all $\theta_j > 0$ (i.e., $\gamma_{2j} \geq \gamma_{1j}$) and one in which $\theta_j < 0$ (i.e., $\gamma_{2j} \leq \gamma_{1j}$). Whether one or the other or both should be used is a specification decision that researchers should make ahead of time depending on context. We will discuss how these decisions may be made subsequently.

Unconstrained Model: The final row shows a relationship that violates all the above constraints. Here, the effect of being a Marine is relatively polarizing as there is a higher probability that Marines are never sad and always sad compared to their college peers. We denote this model as $\mathcal{M}_u$, and there are again $2 \times (J-1)$ free parameters. In contrast to the dominance model, however, there are no constraints on the values that the bounds may take (so long as they order within a condition). Thus, the unconstrained model posits that there is an effect of condition, but it cannot be captured by a single relation, $\mathcal{M}_u\colon\ \theta_j \in \mathbb{R}$.

Priors on Parameters

Our approach here is Bayesian, and in Bayesian analysis priors are needed on parameters. All four models considered here comprise $J-1$ parameters for the mean bounds α_j. We place a normal prior on these means::

$$\alpha_j \sim \text{Normal}(0, b_\alpha),$$

where b_α is a prior variance setting that must be chosen before analysis. The constant shift model, the dominance model, and the unconstrained model comprise one or more difference parameters θ_j. As for the mean bounds, a normal prior is a reasonable choice for θ_j:

$$\theta_j \sim \text{Normal}(0, b_\theta),$$

where b_θ must again be specified before analysis.

Here is some guidance for setting b_α and b_θ: Since bounds are placed on a standard

normal, it makes sense that values of b_α should be around 1.0. Figure 2 shows the marginal prior distribution on ratings probabilities for 5 rating options and for select values of b_α. For $b_\alpha = 1$, middle panel, the marginal priors have the same distribution, centered around .2, for each of the five rating options. Small values of b_α correspond to a belief that extremes are used excessively at the expense of the middle category (left panel); a large value of b_α corresponds to a belief that extremes are used rarely (right panel). The setting $b_\alpha = 1$ is a good default, and it is hard to imagine reasonable settings smaller than 1/10 and larger than 10.

The prior variance on the difference parameters, in contrast, should typically be much smaller than on the mean bounds. As for any difference parameter, however, the exact choice depends on the analyst's expectation and should be determined by substantive, rather than statistical, arguments. For our purposes, we choose a prior variance of $b_\theta = 0.1$, that is, 1/10th of b_α. We address the consequences of this choice and how it affects model comparison results subsequently.

Data Visualization

The four models correspond to the following helpful data visualizations. Much like in signal-detection analysis, the running cumulative proportions become the target for plotting. Table 2 shows the cumulatives for the two scenarios. The usual approach is to plot *receiver operating characteristic curves* (ROCs), and an example for Scenario I is shown in Figure 3A. The levels of constraint are as follows: If the null model holds, the ROC curve traces the diagonal. If the shift model holds, then the typical curve may be observed. The dominance model implies that the points all lie on one or the other side of the diagonal. The unconstrained model implies only that the points increase on the x and y axes, respectively. For analyzing real-world contrasts, it is advantageous to plot the differences across the conditions as in Figure 3B. The advantage here is that it is easier to spot trends because the y axis may be scaled for differences rather than the entire range from 0 to 1. The constraints

now center around the horizontal zero line. The null model corresponds to this line; the shift and dominance model correspond to curves strictly on one side of it; the unconstrained model has no such constraint. Figures 3C and 3D show the ROC and the difference plot for the data in Scenario II.

It is also possible to take inverse-probit transforms of the cumulative proportions before taking differences. In this case, the shift model implies the curve is a straight horizontal line. Researchers who take this approach, however, have to worry about increased variability from the transform at the ends of the range. For this reason, we do not take the transform.

Bayes Factors

We can measure the strength of evidence from the data for the four models using *Bayes factors* (Jeffreys, 1961), which is a measure of how well each model predicted the data before they are observed (Rouder & Morey, 2019). Readers who are new to Bayes factors are invited to consider one of the many tutorials on their use, and perhaps one of the most helpful resources is the recent 2018 *Psychonomic Bulletin & Review* special issue on Bayesian inference (Vandekerckhove, Rouder, & Kruschke, 2018).

There are many approaches to computing Bayes factors. For the models developed here, we use two different approaches as follows: Some models differ in dimensionality. For example, for $J = 5$ response options, there are eight parameters in the unconstrained model, five in the shift model, and four in the null model. Where the models differ by a relatively small number of parameters, we find that the *bridge sampling* approach proposed by Meng and Wong (1996) works well. Gronau et al. (2017) provide a detailed and accessible tutorial on computing Bayes factors with bridge sampling. The approach has been implemented in an R package by Gronau, Singmann, and Wagenmakers (2020), which we use in our work as well.

We follow a different approach to compare models that have the same number of

parameters, namely, the unconstrained and dominance model. The dominance model is more constrained by virtue of the inequalities. Thus, although the models have the same dimensionality, the parameter space for the dominance model is smaller than that for the unconstrained model. In fact, the unconstrained model *encompasses* the dominance model. When models are encompassed, the Bayes factor may be computed by considering the posterior and prior probabilities of the constraint under the unconstrained model (Gelfand, Smith, & Lee, 1992). The Bayes factor between the dominance and the unconstrained model is

$$B_{du} = \frac{Pr(\mathcal{M}_d|\boldsymbol{Y})}{Pr(\mathcal{M}_d)}.$$

The first step is calculating the denominator, or the prior probability that one distribution dominates another. In a blog post, Ossadnik (2020) proved that $Pr(\mathcal{M}_d) = 2/J$. The next step is calculating the numerator. In practice, the computation is surprisingly uncomplicated. We follow the approach discussed in Haaf and Rouder (2017), which is based on the pioneering work of Klugkist, Kato, and Hoijtink (2005). One simply counts the relative frequency of posterior samples under $\mathcal{M}_u$ that satisfy the dominance constraint.

As outlined before, there are two dominance conditions because either distribution could possibly dominate the other. A test of stochastic dominance can be two-sided if there is no prediction about which distribution dominates the other. The Ossadnik result that $Pr(\mathcal{M}_d) = 2/J$ is for this two-sided case. In the case of a directed test, that is, where a researcher *a priori* predicts that one distribution dominates the other and not the reverse, the prior probability of the constraint is $1/2 \times 2/J = 1/J$. The posterior probability is estimated as the relative frequency of posterior samples in the predicted direction only. If stochastic dominance is observed in this predicted direction, the corresponding Bayes factor will yield stronger evidence than in the two-sided case. Thus, if theoretical considerations indicate a dominance relation in a specific direction, the Bayes factor should be calculated accordingly.

Software for Computing Bayes Factors

We created a user-friendly R web applet for analysis. The user inputs the frequency counts in two conditions such as in Table 1. The outputs are Bayes factors for the four models. Additional prior inputs, such as the variabilities b_α and b_θ may be provided as well. The web applet is available at https://martinschnuerch.shinyapps.io/likertBF/; the underlying source code as well as a set of useful R functions are available at https://github.com/mschnuerch/likertBF.

We illustrate this applet with the example data about sadness in Marines and college students, Scenario I. A screenshot of the applet while analyzing the data is shown in Figure 4. Once the data are inputted, we may press "Plot Data," and under "Data Visualization," we may see the diagnostic plots that are shown in Figure 3. Then, to compute Bayes factors, we may press "Start Analysis," and after some time for sampling, the Bayes factors are returned. We may even choose which dominance model we wish by selecting the respective output option via the red settings wheel in the "Results" panel. Let's say *a priori* we may have thought college students would be happier. Because we entered the Marines under Condition 1 and the scale ranges from "never sad" to "always sad", we specify the dominance model as "$2 > 1$". The results, shown in the center panel, clearly indicate that the constant shift model is preferred. Finally, by clicking "Plot MCMC", we can visually inspect MCMC samples from the unconstrained models for α_j and θ_j.

Applications

In this section, we provide two real-world examples of these fine-grained analyses. The first example comes from Collingwood, Lajevardi, and Oskooii (2018) who asked respondents their opinions about controversial policies of the US administration under former president Donald Trump, including the ban on immigration from select Islamic nations and the continuation of the Keystone pipeline project. Collingwood et al. (2018) conducted two survey waves: one when the policy was proposed and the other during implementation. The

observed proportions, sample sizes, and means are shown in Table 3.

The second example comes from the Pew Research Center's *Election News Pathways Project*. Over 11,000 respondents were surveyed about their perception of the Covid-19 pandemic in late March, 2020. We contrast two questions: In one, participants were asked to rate how well President Trump was responding to the pandemic; in the other, they were asked to rate how well their respective state leaders were responding to the pandemic. The observed proportions and sample sizes are shown in the panel labeled *All* in Table 4.

Collingwood et al. (2018) claimed that the Muslim immigration ban became more popular after it was implemented. We use the four models to assess whether there really was an effect, and if so, whether it may be captured with a single relationship as implied by the dominance and shift models. Figure 5, top left, shows the difference in cumulatives. As can be seen, the curve does not cross the zero-line, indicating the plausibility of a single relation. The Bayes factors for the four models are shown in Table 5. As expected, the winning model is the shift model, followed by the one-sided dominance model and the null model. Hence, we conclude that there is (weak) evidence for an effect. The effect is simple and can be reduced to a single relation that seems to be well described as a shift in probit space. The same analysis may be applied to the question about the Keystone pipeline. For these data, the null has a Bayes factor of at least 4-to-1 against any competitor indicating moderate evidence for a lack of an effect of wave on the Likert distribution.

Perhaps the most interesting data are those about leadership in the Covid-19 pandemic. Here, we have strong evidence for an indominant effect. The unconstrained model is preferred by several hundred orders of magnitude to any competitor. Donald Trump is a polarizing figure compared to state leaders. People were more likely to give Donald Trump extreme ratings than state leaders. This polarization may be seen in the difference plot in Figure 5 (bottom right panel). Here, the curve crosses zero, and though the deflection may appear slight, it is highly evidential because the sample sizes are so large. Accordingly, it

makes little sense to discuss whether Donald Trump is viewed as having responded better or worse than local leaders.

The stochastic indominance is easily resolved in this case by conditioning the data on political-party preference. Among those that are Republican, Donald Trump is judged quite well in responding to the crisis; among those that are Democratic, he is judged quite poorly. This partisan divide is not present among state leaders. Thus, when we condition responses on political-party preference, the dominance model in the expected direction wins. It seems that former US president Donald Trump has made even this grave issue of life and death polarizing and partisan.

Sensitivity To Prior Settings

The Bayesian analysis presented here requires the analyst to set the prior variances on mean bounds and effects (b_α, b_θ). Such requirements have given some researchers pause in adopting Bayesian methods. Indeed, it seems reasonable as a starting point to require that if two researchers run the same experiment and obtain the same data, they should reach similar, if not the same, conclusions. Yet, Bayesians acknowledge that priors affect their inference. To harmonize Bayesian inference with the above starting point, some analysts actively seek to minimize these effects by picking likelihoods, prior parametric forms, and heuristic methods of inference so that variation in prior settings have minimal influence (Aitkin, 1991; Gelman, Carlin, Stern, & Rubin, 2004; Kruschke, 2013; Spiegelhalter, Best, Carlin, & van der Linde, 2002). Accordingly, the dependence may be a threat to the validity of Bayesian analysis as it provides more researcher degrees of freedom (Simmons, Nelson, & Simonsohn, 2011).

We reject the starting point above including the view that minimization of prior effects is necessary. The choice of prior settings is important because it affects the models' predictions about data. Therefore, these settings necessarily affect model comparison.

Whatever this effect, it is the degree resulting from the usage of Bayes rule, which in turn mandates that evidence for competing models is the degree to which they improve predictive accuracy.

When different researchers use different priors, they may arrive at different opinions about the data. This variation is not problematic, however, so long as various prior settings are *justifiable*: The variation in results reflects the legitimate diversity of opinion (Rouder, Morey, & Wagenmakers, 2016). When different reasonable prior settings suggest conflicting conclusions, the data simply do not afford the precision to arrive at a clear verdict between the positions.

With this argument as context, we may assess whether reasonable variation in prior settings affect Bayes factor conclusions among the models. In Figure 2, we show that $b_\alpha = 1$ is a good default choice for the prior on α_j, and this choice may be made without undue influence on model comparison results. The prior choice on the difference parameters θ_j is more consequential. For the previous analysis, we specified $b_\theta = 0.1$. For this setting, we consider a range from .02 (1/5th the original setting) to 0.5 (5 times the original setting) to be reasonable. Values of $b_\theta < 0.02$ place excessive weight on extremely small differences between conditions, while values of $b_\theta > 0.5$ place excessive weight on overwhelmingly large differences.

To see how variation in this prior setting affects the Bayes factors, we use a modified version of the Election News Pathways Project data. Unfortunately, with $11,000$ observations, the sample size is quite large to be typical of psychological data. A more typical set would have fewer observations, and so for the purposes here we took the frequencies in Table 4 and divided them by 10. We used the complete data set with both Republicans and Democrats because here we found strong evidence for an indominant effect. Along with these data, we used the subset of Republicans as these showed evidence for a simpler structure, namely, stochastic dominance.

The Bayes factors for the modified data set with the same prior setting as in the previous analysis ($b_\alpha = 1.0$ and $b_\theta = 0.1$) are shown in Table 6. For all the data, the unconstrained model is still preferred over the others. The closest competitor is the dominance model, and the corresponding Bayes factor is approximately 7-to-1. The reason this value is more moderate than that in Table 4 reflects the reduced sample size. When 1/10th the data are used, the winning model for Republicans is the dominance model, preferred over the unconstrained model by a factor of approximately 4-to-1, as in the previous analysis. The question is whether these two values depend heavily on the range of prior settings.

The dependence is shown in Figure 6. Here, Bayes factors of all models against the preferred model within three orders of magnitude (10^{-3}) are displayed. Although the exact figures vary slightly, there is virtually no dependence of Bayes factors across the reasonable range of prior settings. Both for the complete set (left panel) and the subset (right panel), the winning model is preferred over its nearest competitor by a relatively constant amount. Hence, the Bayes factor method provides for evidence that is fairly robust to reasonable variation in prior expectations about data.

Conclusion

Although the use of Likert scales is exceedingly popular, we argue here that researchers have overlooked a defining primitive in analysis (see also Clason & Dormody, 1994). Before calculating means (or other statistics of location) and comparing them, we need to ask whether the ratings distributions can be compared with a single relation. If so, the relationship between two distributions may be captured with simple metrics such as mean differences and t-values. If not, however, common parametric and even nonparametric tests of differences miss the underlying structure and may mislead the analyst.

The statistical models developed herein allow for a more fine-grained analysis of Likert

item data. The null, constant shift, dominance, and unconstrained models provide for a rich description of possible structure in the relationship between two distributions, and strength of evidence from data for them may be stated via Bayes factors. The models as well as the Bayes factor comparisons are straightforward and computationally convenient. We demonstrated their usefulness with two real-world examples and created an easily accessible, user-friendly, freely available web applet for researchers.

Although we think that researchers will benefit from the development herein, there are also limitations. So far, the development only applies to the comparison of two Likert items. Of course, psychologists are often interested in more complicated designs. For example, the data from Collingwood et al. (2018) are panel data in which the same people answered in both waves. We do not take into account any shared variation from the panel design. Moreover, Likert scale analysis is rarely run for a single item across just two levels of a covariate. It is more typical to use multiple items to construct latent scales. And in this case, questions about a shift or stochastic dominance in the data should be addressed at the scale level. Thus, future efforts may be devoted to extending our approach to account for more complex questions and data structures. At this point, the development presented herein constitutes a useful first step toward a more complete framework of meaningful analysis of Likert scales.

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Table 1
Likert scale for hypothetical sadness example.

	Never	Rarely	Sometimes	Often	Always
Scenario I					
Marines	30	25	20	15	10
College Students	20	20	20	20	20
Scenario II					
Marines	40	8	4	8	40
College Students	20	20	20	20	20

Note. Question: How often do you feel sad?

Table 2
Cumulative proportions for hypothetical sadness example.

	Observed Proportions						
	Never	Rarely	Sometimes	Often	Always	N	Mean
Scenario I							
Marines	0.3	0.55	0.75	0.9	1	100	2.5
College Students	0.2	0.4	0.6	0.8	1	100	3
Scenario II							
Marines	0.4	0.48	0.52	0.6	1	100	3
College Students	0.2	0.4	0.6	0.8	1	100	3

Note. Question: How often do you feel sad?

Table 3
Likert Scale Distributions from Collingwood et al. (2018).

	Observed Proportions						
	Strongly Disagree	Disagree	Neutral	Agree	Srongly Agree	N	Mean
Immigration Ban[1]							
First Wave	0.3	0.14	0.14	0.14	0.29	411	2.97
Second Wave	0.4	0.11	0.09	0.16	0.23	311	2.71
Keystone Pipeline[2]							
First Wave	0.42	0.08	0.18	0.13	0.18	409	2.56
Second Wave	0.39	0.14	0.12	0.14	0.2	311	2.61

Note. 1. Agreement with President Trump's executive order restricting immigration from Syria, Iran, Iraq, Libya, Yemen, Somalia, and Sudan. 2. Agreement with President Trump's executive order allowing for the Keystone and Dakota Access Pipelines.

Table 4
Likert Scale Distributions from the Election News Pathway Project.

	Observed Proportions					
	Excellent	Good	Fair	Poor	N	Mean
All						
Trump	0.24	0.25	0.19	0.32	11491	2.6
State Officials	0.21	0.49	0.22	0.08	11432	2.16
Democrats						
Trump	0.04	0.14	0.26	0.56	5937	3.33
State Officials	0.21	0.48	0.23	0.07	5914	2.16
Republicans						
Trump	0.47	0.36	0.11	0.06	5101	1.76
State Officials	0.21	0.52	0.2	0.08	5076	2.14

Note. How would you rate the job each of the following is doing responding to the coronavirus outbreak? A. Donald Trump. B. Your elected state officials.

Table 5
Bayes Factors for Empirical Examples.

	Null	Shift	Dominance	Unconstrained
Immigration Ban	0.29	1.00	0.50	0.14
Keystone Pipeline	1.00	0.25	0.08	0.23
Covid, All	0.00	0.00	0.00	1.00
Covid, Democrats	0.00	0.00	1.00	0.25
Covid, Republicans	0.00	0.00	1.00	0.25

Note. The winning model is assigned a value of 1.00. Bayes factors for all other models are relative to this winning model.

Table 6
Bayes Factors for Modified Election News Pathways Project Data.

	Null	Shift	Dominance	Unconstrained
Covid, All	0.00	0.00	0.14	1.00
Covid, Republicans	0.00	0.00	1.00	0.29

Note. The winning model is assigned a value of 1.00. Bayes factors for all other models are relative to this winning model.

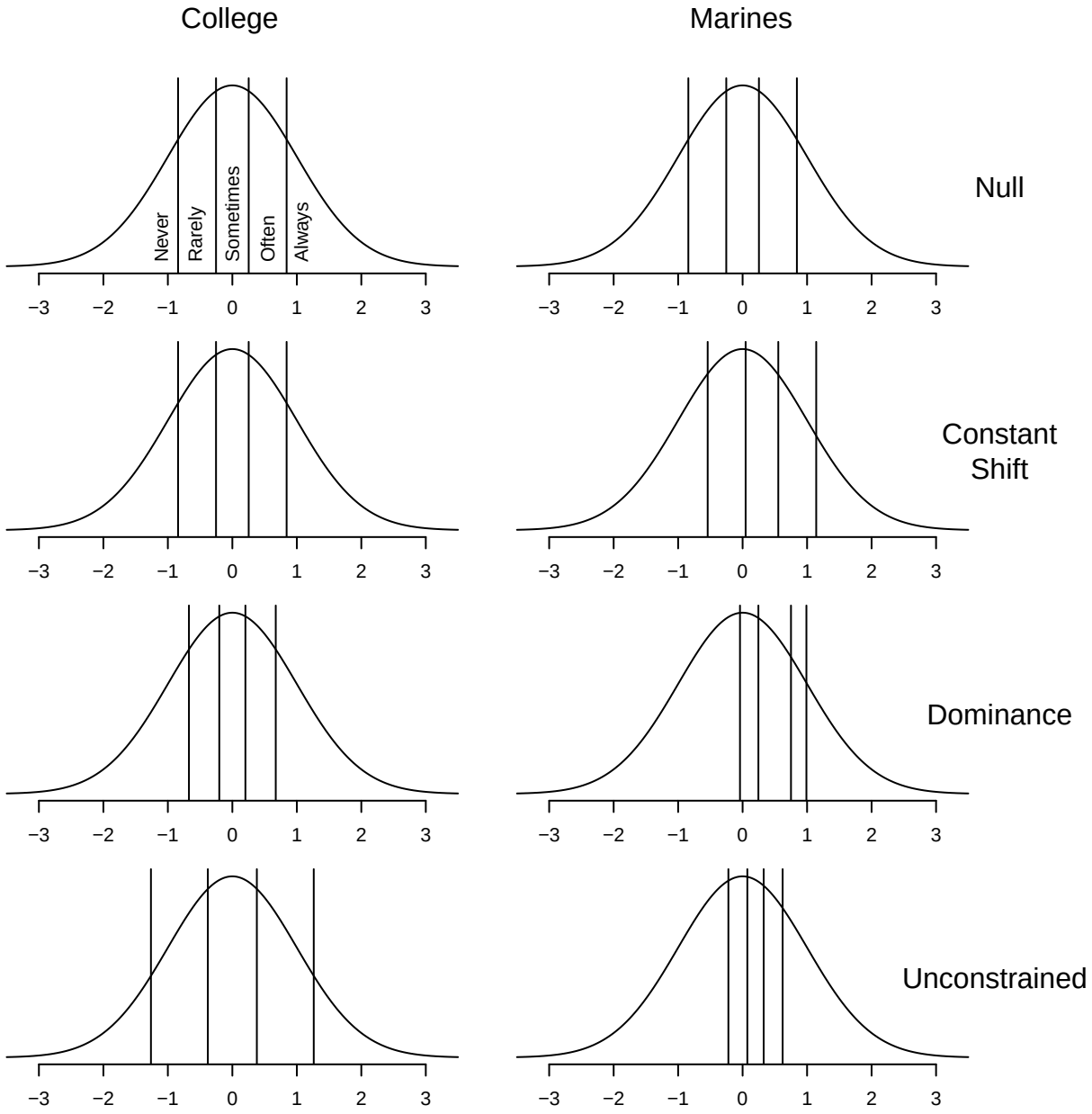

Figure 1. Signal-detection setup with bounds between categories. In this application, the distribution is fixed as a standard normal and the bounds are free parameters. Models, depicted across the rows, capture the relations across conditions (college students vs. Marines).

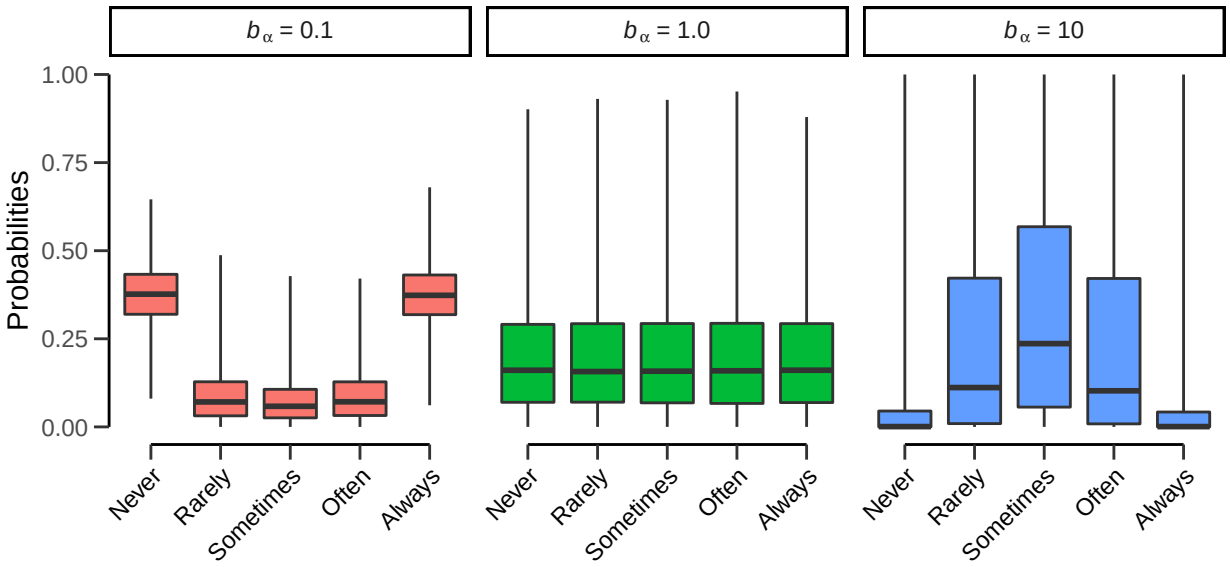

Figure 2. Marginal prior distributions on category probabilities for 5 rating options for select prior variance settings (b_α).

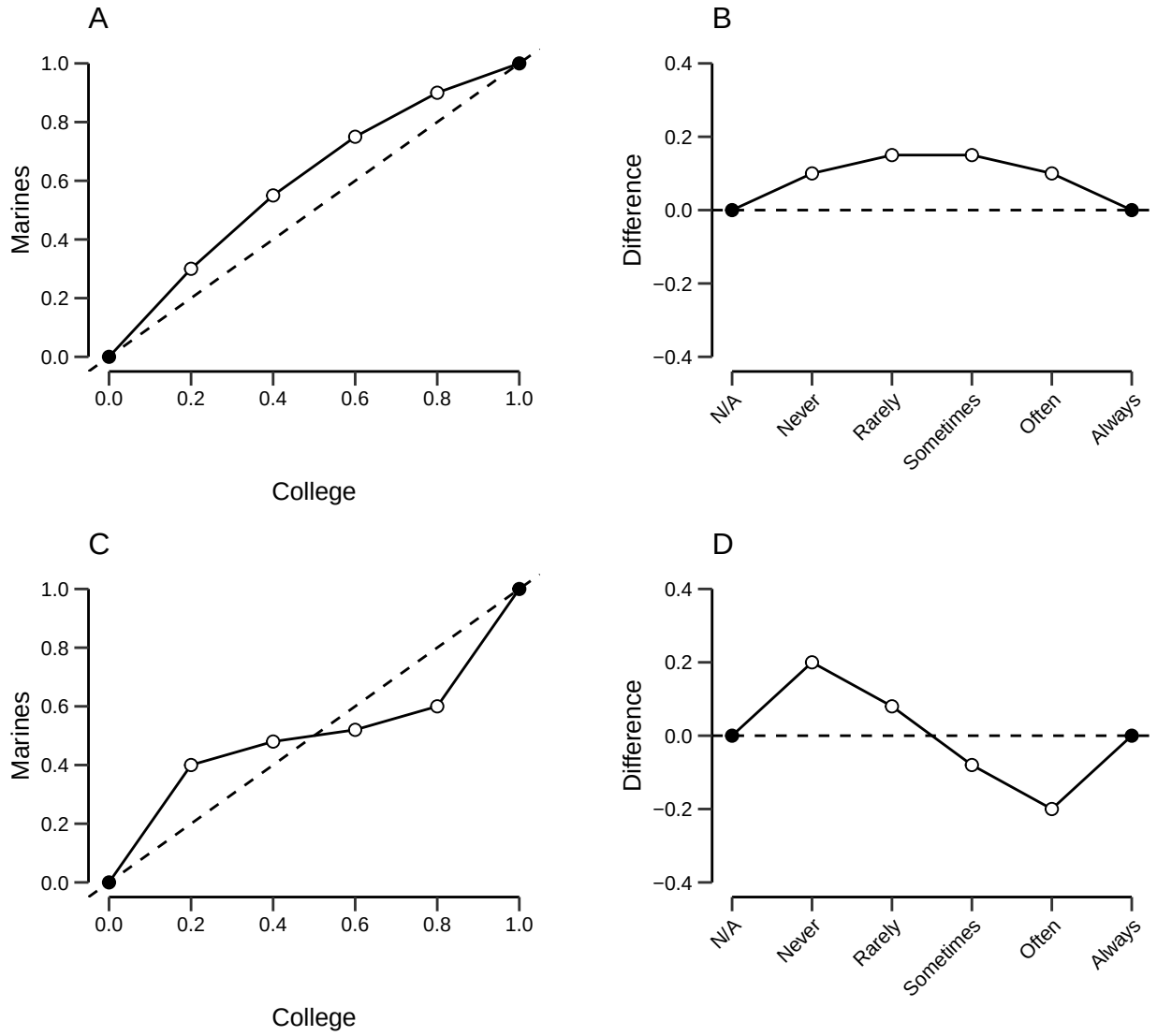

Figure 3. ROC and difference plots for the hypothetical data from Scenarios I and II in Table 1

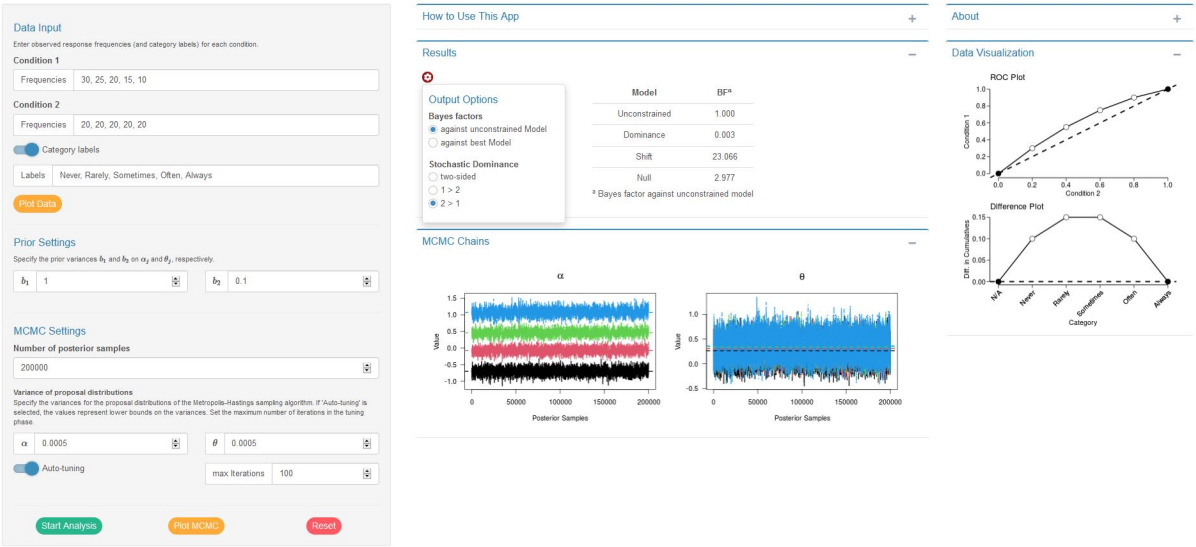

Figure 4. Screenshot of the R web applet for Bayes factor analysis of Likert items. The analyzed data correspond to the hypothetical Scenario I in Table 1.

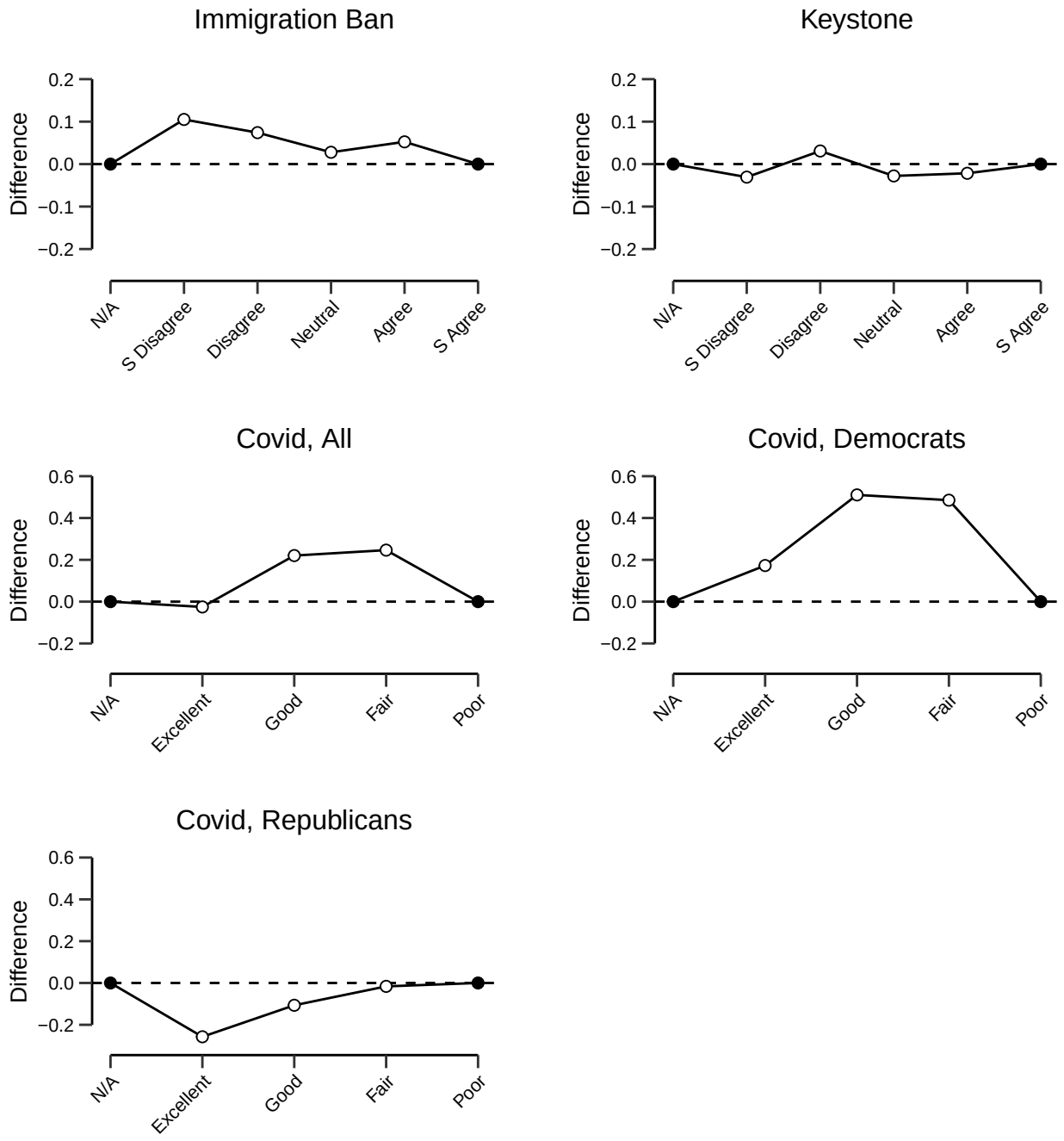

Figure 5. Difference plots for the real-world data in Tables 3 and 4. There is evidence for a constant shift in the top-left panel and evidence for a lack of an effect in the top-right panel. In the middle-left panel, there is strong evidence for an indominant effect, while there is evidence for stochastic dominance in the remaining figures.

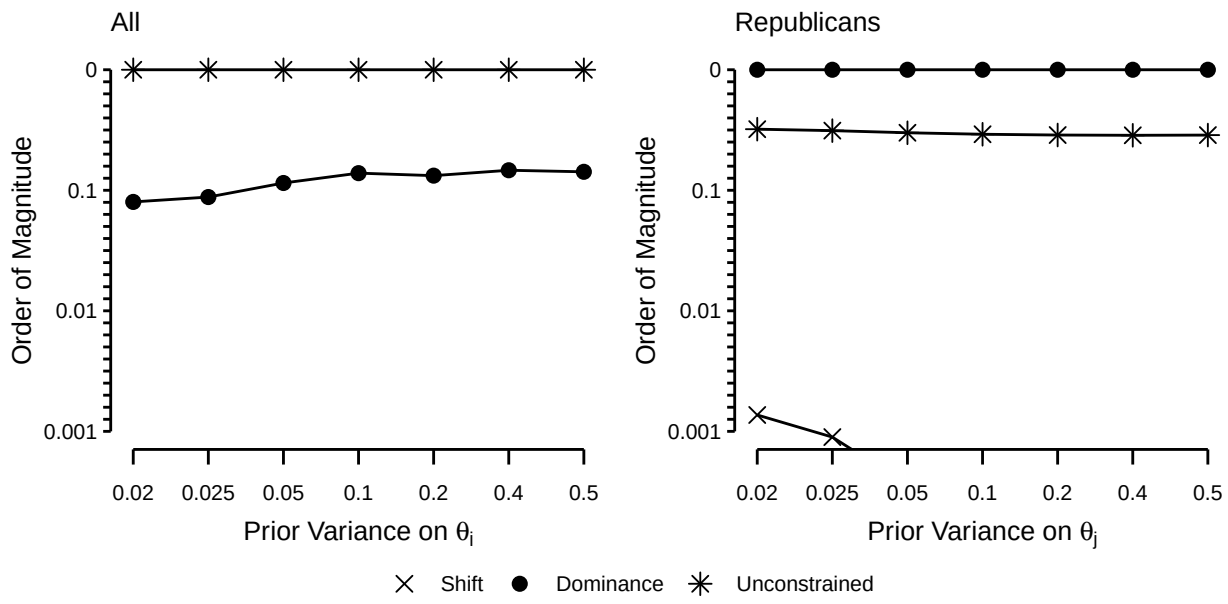

Figure 6. Dependence of Bayes factors on prior variance setting (b_θ) for 1/10th of the Election News Pathways Project data. Only models with Bayes factors within three orders of magnitude (10^{-3}) against the preferred model are shown.